

✓ Oasis Infobyte Internship

Task 1: Exploratory Data Analysis (EDA) on Retail Sales Data

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```
!pip install xlrd
```

Requirement already satisfied: xlrd in /usr/local/lib/python3.12/dist-packages (2.0.2)

✓ 1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

✓ 2. Load the Dataset

```
data = pd.read_excel("/content/sample_-_superstore.xls")
```

✓ 3. Display First Five Rows

```
data.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	...	Postal Code	Region	Product ID
0	1	US-2023-103800	2023-01-03	2023-01-07	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	...	77095	Central	OFF-P/100017
1	2	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-E/100040
2	3	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-L/100032
3	4	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-S/100027
4	5	US-2023-141817	2023-01-05	2023-01-12	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	...	19143	East	OFF-A/100034

5 rows × 21 columns

Observation

- The first five rows provide a quick preview of the dataset.
- The dataset contains customer details, product information, sales, profit, discount, quantity, and shipping details.
- This helps in understanding the structure and contents of the dataset before performing further analysis.

4. Dataset Information

```
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   Row ID                 10194 non-null  int64   
1   Order ID               10194 non-null  object  
2   Order Date             10194 non-null  datetime64[ns]
3   Ship Date              10194 non-null  datetime64[ns]
4   Ship Mode              10194 non-null  object  
5   Customer ID            10194 non-null  object  
6   Customer Name          10194 non-null  object  
7   Segment               10194 non-null  object  
8   Country/Region         10194 non-null  object  
9   City                   10194 non-null  object  
10  State/Province         10194 non-null  object  
11  Postal Code            10194 non-null  object  
12  Region                 10194 non-null  object  
13  Product ID             10194 non-null  object  
14  Category               10194 non-null  object  
15  Sub-Category           10194 non-null  object  
16  Product Name           10194 non-null  object  
17  Sales                  10194 non-null  float64  
18  Quantity               10194 non-null  int64   
19  Discount               10194 non-null  float64  
20  Profit                 10194 non-null  float64  
dtypes: datetime64[ns](2), float64(3), int64(2), object(14)
memory usage: 1.6+ MB
```

Observation

- The dataset contains **10,194 rows** and **21 columns**.
- There are no missing values in any column.
- The dataset contains both numerical and categorical features.
- The data is clean and ready for further analysis.

5. Dataset Shape

```
print("Dataset Shape:", data.shape)
```

```
Dataset Shape: (10194, 21)
```

Observation

- The dataset consists of **10,194 rows** and **21 columns**.
- A large number of records ensures meaningful analysis.
- The dataset has sufficient features to perform comprehensive exploratory data analysis.

6. checking missing value

```
data.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country/Region	0
City	0
State/Province	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

Observation

- The dataset contains **10,194 rows** and **21 columns**.
- There are **no missing values** in any column.
- The dataset contains both numerical and categorical features.
- The data is clean and ready for analysis.

7. checking duplicaticate rows

```
print("Duplicate Rows:", data.duplicated().sum())
```

Duplicate Rows: 0

8. Statistical Summary

```
data.describe()
```

	Row ID	Order Date	Ship Date	Sales	Quantity	Discount	Profit
count	10194.000000	10194	10194	10194.000000	10194.000000	10194.000000	10194.000000
mean	5097.500000	2025-04-29 11:48:25.002942720	2025-05-03 10:52:45.626839296	228.225854	3.791838	0.155385	28.673417
min	1.000000	2023-01-03 00:00:00	2023-01-07 00:00:00	0.444000	1.000000	0.000000	-6599.978000
25%	2549.250000	2024-05-14 00:00:00	2024-05-19 00:00:00	17.220000	2.000000	0.000000	1.760800
50%	5097.500000	2025-06-25 00:00:00	2025-06-28 00:00:00	53.910000	3.000000	0.200000	8.690000
75%	7645.750000	2026-05-14 00:00:00	2026-05-18 00:00:00	209.500000	5.000000	0.200000	29.297925
max	10194.000000	2026-12-30 00:00:00	2027-01-05 00:00:00	22638.480000	14.000000	0.800000	8399.976000

Observation

- The dataset contains **10,194 records**.
- The average sales per order are **228.23**.
- The average profit per order is **28.67**.
- The maximum sales value is **22,638.48**, while the minimum sales value is **0.44**.
- The minimum profit is **-6599.98**, showing that some orders resulted in heavy losses.
- The discount ranges from **0% to 80%**.

Business Insight

The company should investigate orders with very high losses and review its discount strategy to improve profitability.

9. Total Sales

```
print("Total Sales:", data["Sales"].sum())
```

```
Total Sales: 2326534.3543
```

Observation

This value represents the total revenue generated from all orders in the dataset.

10. Total Profit

```
print("Total Profit:", data["Profit"].sum())
```

```
Total Profit: 292296.81459999999
```

Observation

The total profit indicates the overall earnings after deducting costs from sales.

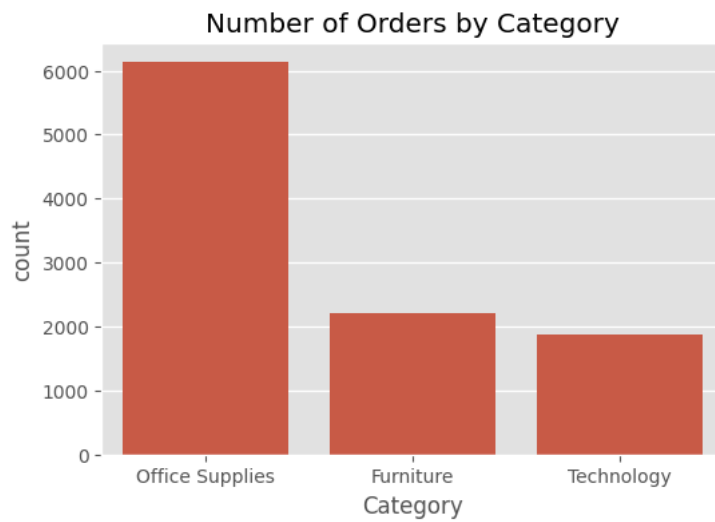
11. Category Distribution

```
data["Category"].value_counts()
```

	count
Category	
Office Supplies	6128
Furniture	2201
Technology	1865

```
dtype: int64
```

```
plt.figure(figsize=(6,4))
sns.countplot(x="Category", data=data)
plt.title("Number of Orders by Category")
plt.show()
```



Observation

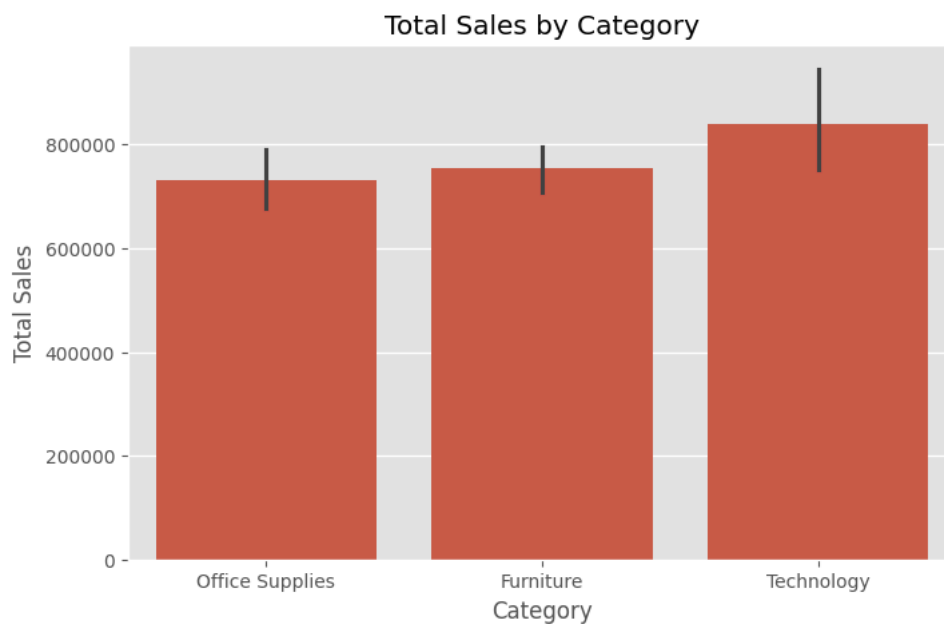
- Office Supplies has the highest number of orders.
- Furniture is the second most ordered category.
- Technology has the lowest number of orders.

Business Insight

Office Supplies are purchased frequently, making them an important source of regular sales. Maintaining sufficient inventory for this category can help meet customer demand.

✓ 12. Sales by Category

```
plt.figure(figsize=(8,5))
sns.barplot(x="Category", y="Sales", data=data, estimator=sum)
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```



Observation

- Technology generates the highest sales revenue.
- Furniture contributes moderate sales.
- Office Supplies have many orders but lower total sales than Technology.

Business Insight

Technology products contribute significantly to revenue. The company should continue promoting this category while identifying ways to increase Furniture sales.

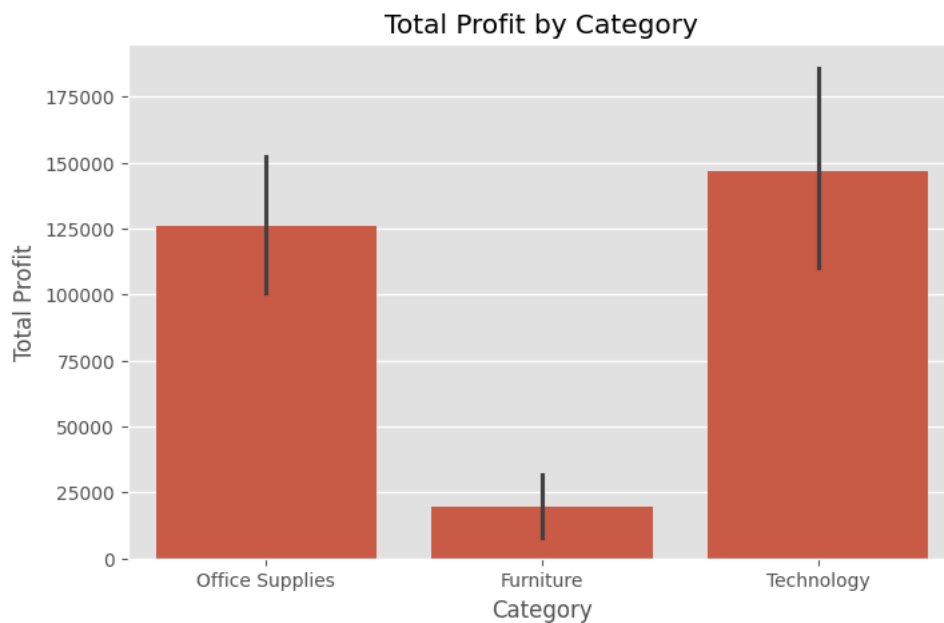
✓ 13. Profit by Category

```
plt.figure(figsize=(8,5))

sns.barplot(x="Category", y="Profit", data=data, estimator=sum)

plt.title("Total Profit by Category")
plt.xlabel("Category")
plt.ylabel("Total Profit")

plt.show()
```



Observation

- Technology category generates the highest total profit.
- Office Supplies is the second most profitable category.
- Furniture contributes the lowest profit among all categories.

Business Insight

Technology products are the most profitable for the company. The Furniture category should be analyzed to improve profitability by reducing discounts or optimizing pricing.

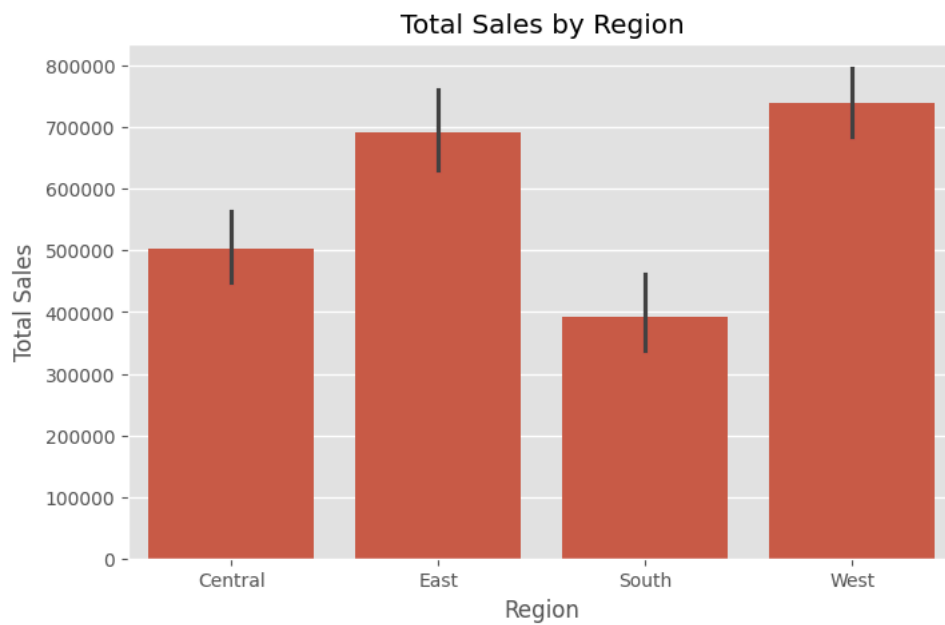
✓ 14. Sales by Region

```
plt.figure(figsize=(8,5))

sns.barplot(x="Region", y="Sales", data=data, estimator=sum)

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
```

```
plt.show()
```



Observation

- West region records the highest total sales.
- East

Business Insight

The West region is the highest revenue-generating region and its successful strategies can be adopted in other regions. The South region requires targeted marketing campaigns and sales strategies to improve its overall performance.

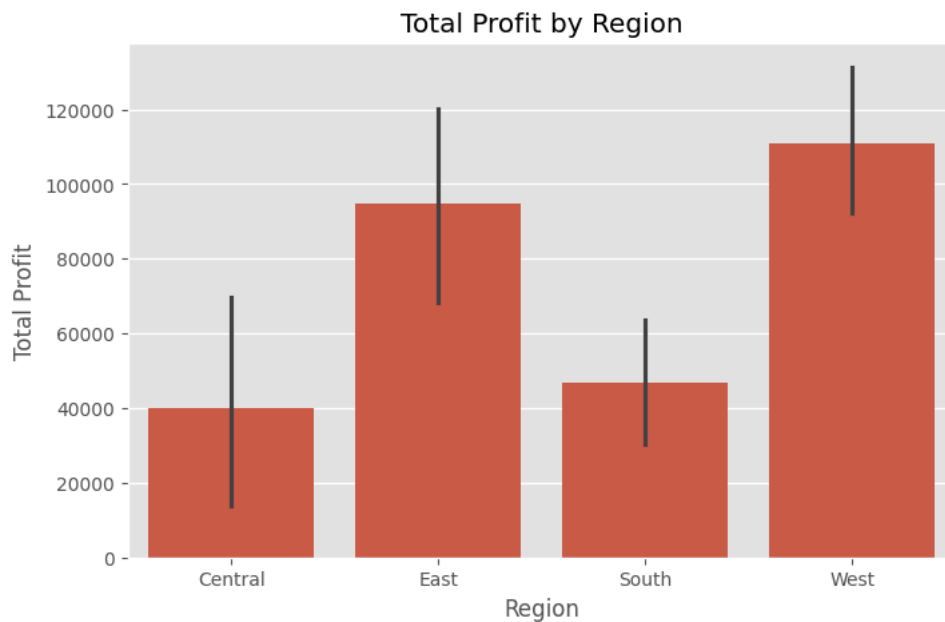
✓ 15. Profit by Region

```
plt.figure(figsize=(8,5))

sns.barplot(x="Region", y="Profit", data=data, estimator=sum)

plt.title("Total Profit by Region")
plt.xlabel("Region")
plt.ylabel("Total Profit")

plt.show()
```



Observation

- Some regions are more profitable than others.
- High sales do not always result in high profits.
- Regional profitability differs due to factors like discounts and product mix.

Business Insight

Management should focus on improving profit margins in regions where sales are high but profits are comparatively low.

✓ 16. Monthly Sales Trend

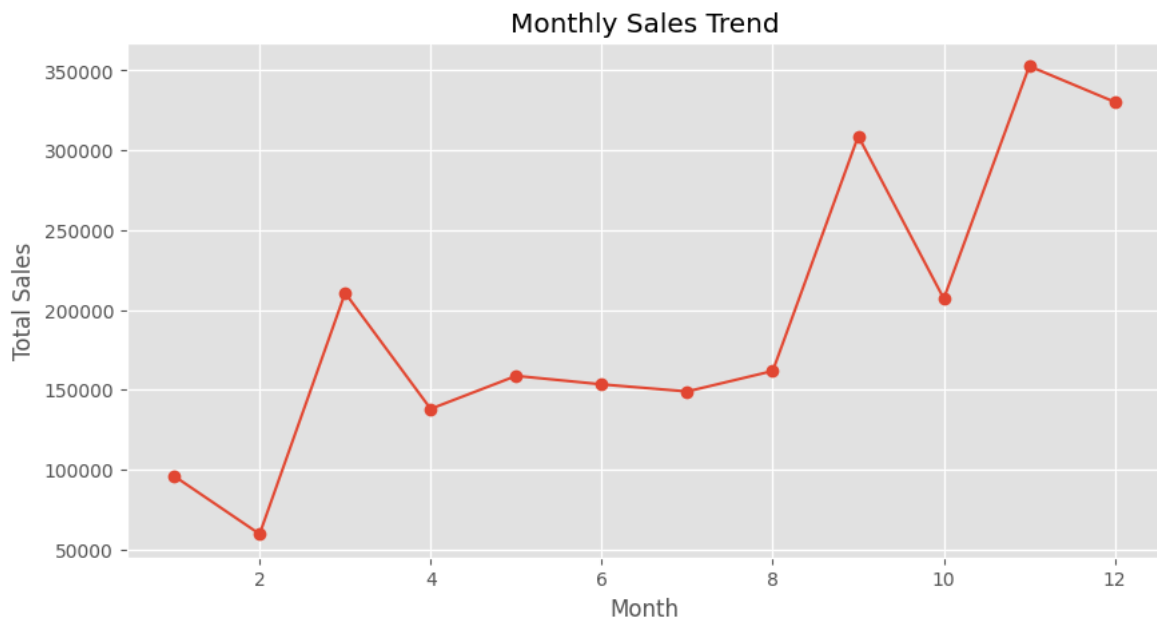
```
monthly_sales = data.groupby(data["Order Date"].dt.month)["Sales"].sum()

plt.figure(figsize=(10,5))

monthly_sales.plot(marker="o")

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.grid(True)

plt.show()
```

Observation

- Monthly sales fluctuate throughout the year.
- Some months experience higher sales than others.
- This indicates the presence of seasonal demand.

Business Insight

The company can plan promotions and inventory based on high-demand months to maximize sales and customer satisfaction.

✓ 17. Top 10 Best Selling Products

```
top_products = data.groupby("Product Name")["Sales"].sum().sort_values(ascending=False).head(10)
```

```
top_products
```

	Sales
Product Name	
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

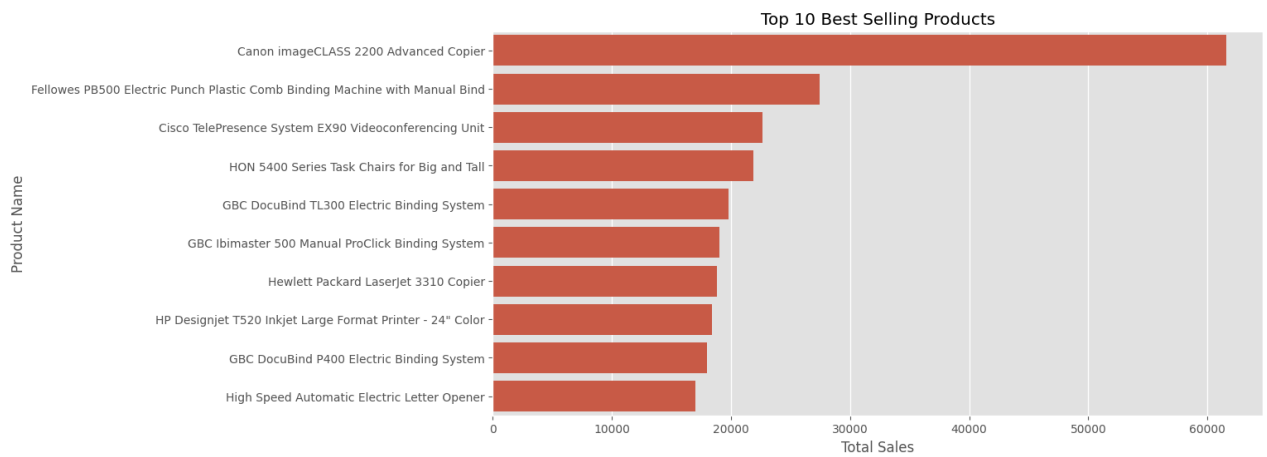
```
dtype: float64
```

```
plt.figure(figsize=(12,6))

sns.barplot(x=top_products.values,
            y=top_products.index)

plt.title("Top 10 Best Selling Products")
plt.xlabel("Total Sales")
plt.ylabel("Product Name")

plt.show()
```



Observation

- The top 10 products contribute significantly to the overall sales.
- A few products generate much higher sales than the remaining products.
- Sales are concentrated among the best-selling products.

Business Insight

The company should ensure adequate inventory for the best-selling products and use targeted marketing to further increase their sales.

✓ 18. Top 10 Most Profitable Products

```
top_profit = data.groupby("Product Name")["Profit"].sum().sort_values(ascending=False).head(10)
```

```
top_profit
```

	Profit
Product Name	
Canon imageCLASS 2200 Advanced Copier	25199.9280
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	7753.0390
Hewlett Packard LaserJet 3310 Copier	6983.8836
Canon PC1060 Personal Laser Copier	4570.9347
HP Designjet T520 Inkjet Large Format Printer - 24" Color	4094.9766
Ativa V4110MDD Micro-Cut Shredder	3772.9461
3D Systems Cube Printer, 2nd Generation, Magenta	3717.9714
Plantronics Savi W720 Multi-Device Wireless Headset System	3696.2820
Ibico EPK-21 Electric Binding System	3345.2823
Zebra ZM400 Thermal Label Printer	3343.5360

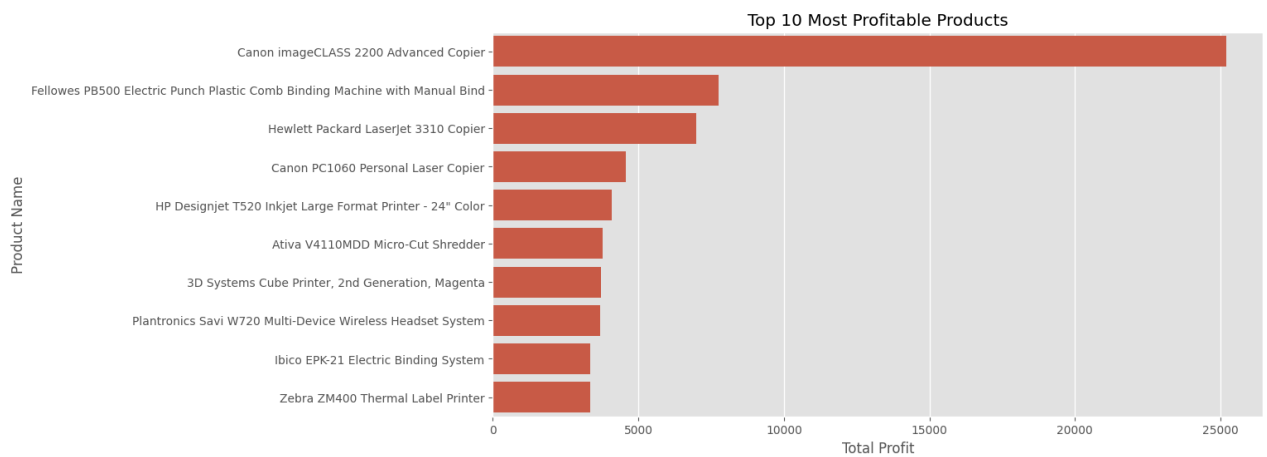
```
dtype: float64
```

```
plt.figure(figsize=(12,6))

sns.barplot(x=top_profit.values,
            y=top_profit.index)

plt.title("Top 10 Most Profitable Products")
plt.xlabel("Total Profit")
plt.ylabel("Product Name")
```

```
plt.show()
```



Observation

- A small number of products contribute the highest profits.
- Profit generation is concentrated among a few products.
- These products are the company's strongest profit drivers.

19. Sales by Customer Segment

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(x="Segment", y="Sales", data=data, estimator=sum)
```

```
plt.title("Total Sales by Customer Segment") plt.xlabel("Customer Segment") plt.ylabel("Total Sales")
```

```
plt.show()
```

Observation

- Sales vary across different customer segments.
- Some customer segments contribute more revenue than others.
- Customer purchasing behavior differs among segments.

Business Insight

The company should focus on high-performing customer segments while designing promotional strategies to improve sales among lower-performing segments.

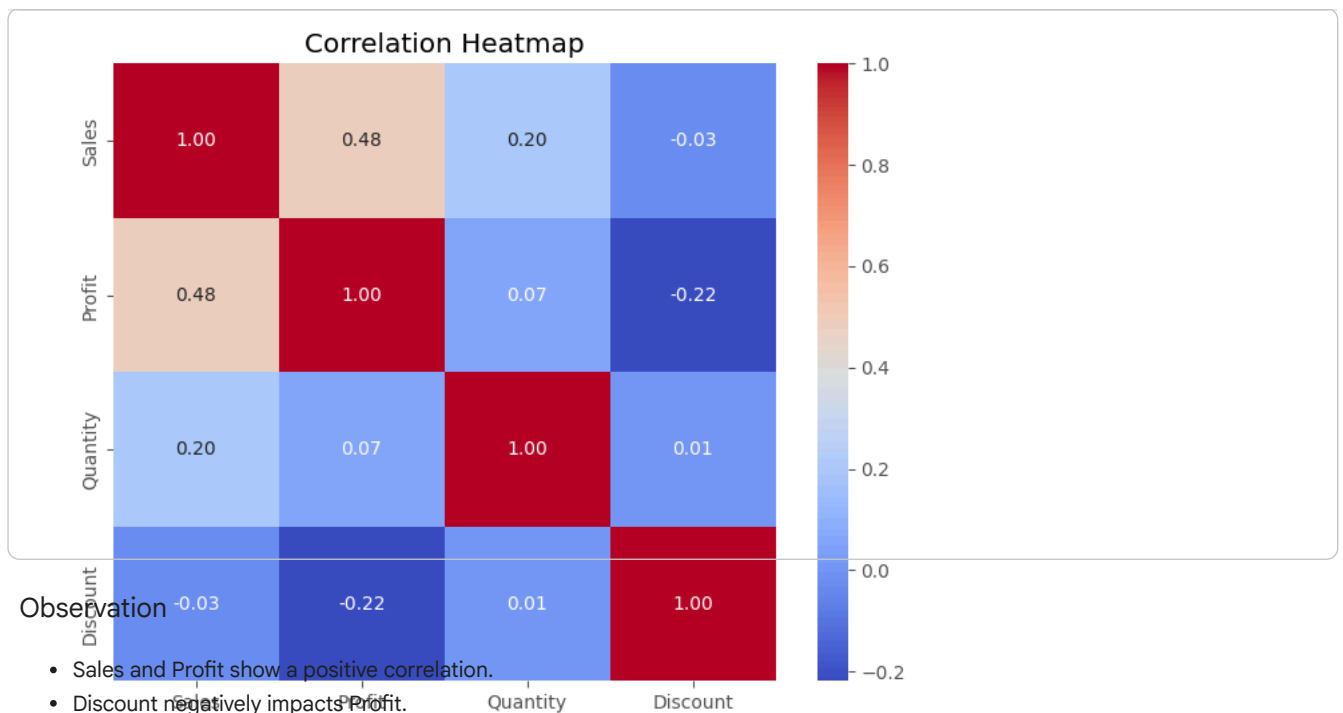
20. Correlation Heatmap

```
plt.figure(figsize=(8,6))
```

```
sns.heatmap(
    data[["Sales", "Profit", "Quantity", "Discount"]].corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```



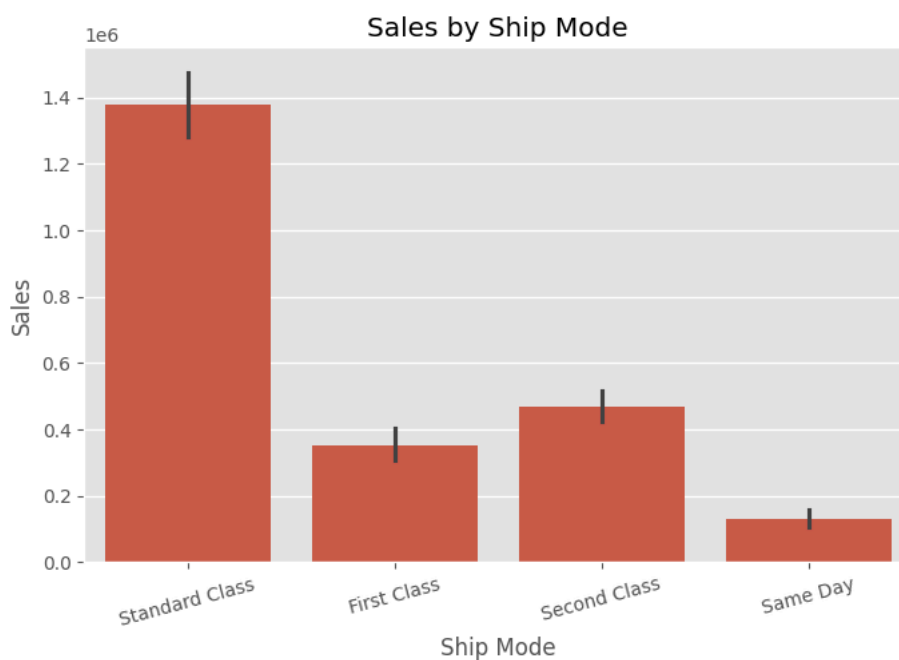
Double-click (or enter) to edit

Business Insight

The company should carefully manage discount policies because higher discounts tend to reduce profitability. Increasing sales while maintaining healthy profit margins will improve overall business performance.

21. Sales by Ship Mode

```
plt.figure(figsize=(8,5))
sns.barplot(x="Ship Mode", y="Sales", data=data, estimator=sum)
plt.title("Sales by Ship Mode")
plt.xticks(rotation=15)
plt.show()
```



Observation

Different shipping modes contribute differently to total sales

